

RESEARCH ARTICLE | OCTOBER 15 1999

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Band gap narrowing in *n*-type and *p*-type 3C-, 2H-, 4H-, 6H-SiC, and Si

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(Received 20 April 1999; accepted for publication 13 July 1999)

Doping-induced energy shifts of the conduction band minimum and the valence band maximum have been calculated for *n*-type and *p*-type 3C-, 2H-, 4H-, 6H-SiC, and Si. The narrowing of the fundamental band gap and of the optical band gap are presented as functions of ionized impurity concentration. The calculations go beyond the common parabolic treatments of the ground state energy dispersion by using energy dispersion and overlap integrals from band structure calculations. The nonparabolic valence band curvatures influence strongly the energy shifts especially in *p*-type materials. The utilized method is based on a zero-temperature Green's function formalism within the random phase approximation with local field correction according to Hubbard. We have parametrized the shifts of the conduction and the valence bands and made comparisons with recently published results from a semi-empirical model. © 1999 American Institute of Physics.
[S0021-8979(99)09020-9]

I. INTRODUCTION

Whereas Si is the most exploited semiconductor, SiC is one of the most promising materials for high power, high temperature, and high frequency applications. SiC-based devices are under development and in order to properly design the devices, simulations of the electrical properties play an important role. For instance, there have been simulations of the electron and hole transport in SiC for both bulk materials and devices.¹⁻³ In these investigations an accurate description of the electronic band structure is crucial, but routinely the band structure of the intrinsic material is used. However, doping can strongly modify the band structure which can be explained by the following:

In an undoped semiconductor the Hamiltonian $\hat{H}_0$ describes the interactions between the ions and the electrons of the intrinsic crystal. The eigenvalue problem of $\hat{H}_0$ is normally solved by a band structure calculation, resulting in the single-electron energies $E_j^0(\mathbf{k})$ for the electron states with wave vector $\mathbf{k}$ in the *j*th energy band. In the ground state at zero temperature, all electron states in the valence bands are occupied by electrons and all electron states in the conduction bands are unoccupied and the lowest energy needed to promote one electron from the valence bands to the conduction bands is the fundamental band gap E_g^0 .

In an *n*-type semiconductor the crystal also consists of donor atoms which can be thermally ionized, thereby giving rise to a "gas" of free electrons in the conduction band. The crystal electrons interact strongly with the electron gas and with the ionized donors; interactions with the neutral nonion-

ized donors are normally neglected. Hence, the Hamiltonian $\hat{H}$ contains interactions (i) between the electron gas and the ionized donors and (ii) between the electrons within the electron gas. There are also interactions (iii) between the valence band electrons (or alternatively, the holes), and the ionized donors and (iv) between the valence band electrons and the electron gas. If the material is polar there are also (v) interactions between the electrons and the optical phonons. The additional interaction terms in $\hat{H}$ are commonly treated as a perturbation to $\hat{H}_0$, that is, $\hat{H} = \hat{H}_0 + \hat{H}_1$ where $\hat{H}_1$ is the perturbation Hamiltonian describing the additional interactions. The eigenvalue problem of $\hat{H}$ results in the single-electron energies $E_j(\mathbf{k})$ which differ from the corresponding energies of the unperturbed eigenvalue problem. The energy shift $E_j(\mathbf{k}) - E_j^0(\mathbf{k}) = \text{Re}[\hbar \Sigma_j(\mathbf{k}, E_j^0(\mathbf{k})/\hbar)]$ of a single-electron energy is the real part of the *self-energy* of the electron state. Electron states in both the conduction and the valence bands are affected by the additional interactions and, normally, the self-energy of a conduction band state is negative and the self-energy of a valence band state is positive, thereby giving a smaller energy band gap E_g . The earlier description for *n*-type materials has a direct analogue for *p*-type materials.

In the present work, the energy shifts of the conduction band minimum and of the valence band maximum have been calculated as functions of ionized impurity concentration for both *n*-type and *p*-type 3C-, 2H-, 4H-, 6H-SiC, and Si. The self-energies used have been worked out within the random phase approximation (RPA) with the local field correction of Hubbard in Refs. 4 and 5. A simple approach to incorporate the electronic structure of the unperturbed crystal [that is, $E_j^0(\mathbf{k})$] is to use parabolic energy dispersions. Also overlap

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integrals between two electron states, contained in the descriptions of various scattering mechanisms involved in the interactions, can be parametrized within a parabolic approximation. In this work, the “true” energy dispersions and overlap integrals (i.e., the energy dispersions and the overlap integrals from a band structure calculation of C. Persson and U. Lindefelt)^{4,6,7} have been taken into account. It is shown that the nonparabolicity of the valence band curvatures affects the calculated self-energies of the valence band maximum. In *p*-type Si the agreement between calculated and measured values of the band gap narrowing are considerably improved when the nonparabolicities are included. Also the double-well structure of the lowest conduction band in 6H-SiC has a strong influence on the self-energies, especially for high ionized impurity concentrations.

Throughout the article centimeter-gram-second units have been employed. The labeling of symmetry points and symmetry lines of the Brillouin zones (BZ) is the same as in Ref. 6. The k_z direction is defined as the longitudinal direction [which is along the $\Gamma X(\Gamma A)$ direction in the cubic (hexagonal) materials] and the k_{xy} plane is the transverse direction perpendicular to the k_z direction. We have taken into account the lowest conduction band ($c1$) and the three uppermost valence bands ($v1$, $v2$, and $v3$, where $v1$ is the uppermost band). In cubic materials $v1$, $v2$, and $v3$ correspond to heavy-hole, light-hole, and spin-orbit split-off bands, respectively.

II. COMPUTATIONAL METHOD

The employed method for calculating the self-energy is within the RPA with a local field correction of Hubbard.⁸ We have considered exchange and correlation energies due to electron–electron interactions and, moreover, energies from electron–impurity ion and electron–optical phonon interactions. A zero-temperature formalism is used and temperature enters our model only through its capability to ionize the dopant atoms below the critical concentration for the Mott transition. Above the critical concentration the impurities are fully ionized even at zero absolute temperature. The concentration for Mott transition is about $3.5 \times 10^{18} \text{ cm}^{-3}$ in P-doped Si⁹ and estimated to be $3.5 \times 10^{18} \text{ cm}^{-3}$ in N-doped 3C-SiC and in the range $0.6\text{--}2.6 \times 10^{19} \text{ cm}^{-3}$ in N-doped 4H- and 6H-SiC.⁴ Interactions with the nonionized impurity atoms are neglected. The distribution of the impurity ions have been shown to be of major importance for the results^{10,11} and the present calculations are presuming a random distribution of impurities.

The Coulomb potential $v(\mathbf{q})$ of the electrons and of the ions are screened by virtual transitions across the fundamental band gap. By interpreting the electron–optical phonon interaction as an additional screening of $v(\mathbf{q})$, the dielectric function κ of the intrinsic crystal (the lattice dielectric function) is becoming frequency dependent. The screened Coulomb potential $v(\mathbf{q})/\kappa(\omega) = 4\pi e^2/[\mathbf{q}^2 \kappa(\omega)]$ comprises thereby not only the virtual transitions across the band gap, but also all electron–optical phonon scattering.

The self-energies in Sec. II A have been derived in detail in Refs. 4 and 5, starting from the perturbation Hamiltonian

and the total energy of the electron gas. The resulting self-energy expressions are developed for *n*-type materials, but the corresponding expressions for *p*-type materials can easily be obtained by treating the holes as particles and the electrons as antiparticles. This implies that the self-energies $\hbar \Sigma_j(\mathbf{k}, E_j^0(\mathbf{k})/\hbar)$ should be replaced by $-\hbar \Sigma_j(\mathbf{k}, -E_j^0(\mathbf{k})/\hbar)$.

A. Self-energies

The self-energies are expressed in terms of the unperturbed, time-ordered electron Green’s function $G_j^0(\mathbf{k}, \omega)$ and of the dielectric function $\tilde{\epsilon}(\mathbf{q}, \omega)$ of the electron gas (given in Sec. II B). The unperturbed Green’s function describes the propagation of a noninteracting electron in the crystal and $\tilde{\epsilon}(\mathbf{q}, \omega)$ describes the response of the electron gas due to a perturbation in the electron distribution. The unperturbed Green’s function is defined as

$$G_j^0(\mathbf{k}, \omega) = \frac{\eta_j(\mathbf{k})}{\omega - \xi_j^0(\mathbf{k}) - i\delta} + \frac{1 - \eta_j(\mathbf{k})}{\omega - \xi_j^0(\mathbf{k}) + i\delta}, \quad (1a)$$

$$\xi_j^0(\mathbf{k}) = E_j^0(\mathbf{k})/\hbar, \quad (1b)$$

where δ is an infinitesimal, positive number and $\eta_j(\mathbf{k})$ is one if the state $|\mathbf{k}, j\rangle$ is populated and zero otherwise. If the cell-periodic parts $u_{j\sigma}^0(\mathbf{k}, \mathbf{r})$ of the Bloch functions are assumed to be slowly varying functions of $\mathbf{r}$, then scatterings due to the electron–electron interaction $\hat{V}_{ee}$ can be approximated by

$$\langle \mathbf{k}, j; \mathbf{k}', j' | \hat{V}_{ee} | \mathbf{k}', j'; \mathbf{k}, j \rangle \approx \frac{v(\mathbf{q})}{\kappa(\omega)} \times \Lambda_{j,j'}(\mathbf{k}, \mathbf{k}'), \quad (2a)$$

$$\Lambda_{j,j'}(\mathbf{k}, \mathbf{k}') = \frac{1}{2} \sum_{\sigma, \sigma'=\uparrow, \downarrow} \left| \int d\mathbf{r} u_{j\sigma}^0(\mathbf{k}, \mathbf{r}) u_{j'\sigma'}^0(\mathbf{k}', \mathbf{r}) \right|^2, \quad (2b)$$

where $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ and $\Lambda_{j,j'}(\mathbf{k}, \mathbf{k}')$ is the spin-independent overlap integral.

The Hartree–Fock exchange interaction in the electron gas is defined as the electron–electron interaction caused by the Pauli principle, that is, electrons with equal spin avoid each other. Using the approximation of Eq. (2) the exchange part of the self-energy in the electron gas is^{4,5}

$$\begin{aligned} \hbar \Sigma_j^x(\mathbf{k}, \xi_j^0(\mathbf{k})) = & - \int \frac{d\mathbf{q}}{(2\pi)^3} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi i} \frac{v(\mathbf{q})}{\kappa(\omega)} \\ & \times \sum_{j'} \left\{ \Lambda_{j,j'}(\mathbf{k}, \mathbf{k}') G_{j'}^0(\mathbf{k}', \omega + \xi_j^0(\mathbf{k})) \right. \\ & + \frac{\Lambda_{j,j'}(\mathbf{k}, \mathbf{k}')}{2} \left(\frac{1}{\omega + [\xi_{j'}^0(\mathbf{k}') - \xi_j^0(\mathbf{k})] - i\delta} \right. \\ & \left. \left. - \frac{1}{\omega - [\xi_{j'}^0(\mathbf{k}') - \xi_j^0(\mathbf{k})] + i\delta} \right) \right\}, \quad (3) \end{aligned}$$

where the electrostatic self-interaction of the electron (the second term) has been subtracted. In the Hartree–Fock equation, the direct Coulomb term of the electron repulsion is cancelled by treating the impurity ions as a uniform distribu-

tion of positive charges. The remaining Coulomb repulsion of the electrons is ascribed to the correlation interaction. The correlation self-energy is^{4,5}

$$\hbar \Sigma_j^c(\mathbf{k}, \xi_j^0(\mathbf{k})) = - \int \frac{d\mathbf{q}}{(2\pi)^3} \int_{-\infty}^{\infty} \frac{d\omega}{2\pi i} \frac{v(\mathbf{q})}{\kappa(\omega)} \left(\frac{1}{\tilde{\epsilon}(\mathbf{q}, \omega)} - 1 \right) \times \sum_{j'} \Lambda_{j,j'}(\mathbf{k}, \mathbf{k}') G_{j'}^0(\mathbf{k}', \omega + \xi_j^0(\mathbf{k})). \quad (4)$$

Finally, the self-energy from the electron-impurity ion interaction, presuming a random distribution of the ions with infinitely large masses, is obtained by^{4,5}

$$\hbar \Sigma_j^{\text{ion}}(\mathbf{k}, \xi_j^0(\mathbf{k})) = \frac{N_D^+}{\hbar} \int \frac{d\mathbf{q}}{(2\pi)^3} \left(\frac{v(\mathbf{q})}{\kappa_0 \tilde{\epsilon}(\mathbf{q}, 0)} \right)^2 \times \sum_{j'} \Lambda_{j,j'}(\mathbf{k}, \mathbf{k}') G_{j'}^0(\mathbf{k}', \xi_j^0(\mathbf{k})). \quad (5)$$

The electron-impurity ion interaction describes the relaxation of the electrons around the impurity ions and the resulting inhomogeneity of the electron gas is thereby comprised here. Since a part of the electron-impurity ion interaction already has been taken into account in the Hartree–Fock exchange energy, Eq. (5) describes the self-energy from the difference in the electron density between a system with uniform distribution of the ion charges and a system of point-like ions.

In measurements of the fundamental band gap, an electron and a hole are created in the conduction band minimum and in the valence band maximum, respectively. The exchange, the correlation, and the electron-impurity ion interactions will all contribute to lower the energies of the electron and of the hole. This implies that the conduction band minimum will shift downwards and the valence band maximum will shift upwards, due to these interactions, thereby reducing the band gap.

We want to point out that if a general eigenstate (i.e., not necessarily at the conduction band minimum or the valence band maximum) is considered or if one uses calculated value of the fundamental band gap, an additional polaron self-energy term has to be included.⁴

B. Dielectric functions

The frequency dependent lattice dielectric function $\kappa(\omega)$ is expressed in terms of the static (κ_0) and high-frequency (κ_∞) limits. The phonons involved in the screening are the longitudinal optical phonons with long wave lengths, approximated to have constant phonon dispersion $\hbar \omega_{\text{LO}}$ ⁵

$$\frac{1}{\kappa(\omega)} = \frac{1}{\kappa_\infty} - \frac{(\kappa_0 - \kappa_\infty) \omega_{\text{LO}}}{2 \kappa_0 \kappa_\infty} \left\{ \frac{1}{\omega + \omega_{\text{LO}} - i\delta} - \frac{1}{\omega - \omega_{\text{LO}} + i\delta} \right\}. \quad (6)$$

We have employed experimental values^{12,13} for the lattice dielectric function. For 3C-(6H-) SiC $\kappa_0 = 9.72(9.78)$, $\kappa_\infty = 6.52(6.58)$, and $\hbar \omega_{\text{LO}} = 0.121(0.120)$ eV. The dielectric constants for 6H-SiC are the geometric average values of the transverse and the longitudinal components: $\kappa_{0,\perp} = 9.66$,

TABLE I. Experimental values of the low temperature fundamental band gaps E_g^0 for undoped material (Refs. 12 and 13). The position $\mathbf{k}_m$ of the ν equivalent minima and the effective electron masses in the lowest conduction band are from a band structure calculation (Refs. 6 and 7).

	Si	3C-SiC	2H-SiC	4H-SiC	6H-SiC
E_g^0 (eV)	1.17	2.40	3.33	3.29	3.10
$\mathbf{k}_m$	Δ	X	K	M	Δ
ν	6	3	2	3	6
$m_x(m_0)$	0.19	0.23	0.43	0.57	0.75
$m_y(m_0)$	0.19	0.23	0.43	0.28	0.24
$m_z(m_0)$	0.95	0.68	0.26	0.31	1.83

$\kappa_{0,\parallel} = 10.03$, $\kappa_{\infty,\perp} = 6.52$, and $\kappa_{\infty,\parallel} = 6.70$. Furthermore, 2H- and 4H-SiC are presumed to have the same lattice dielectric function as 6H-SiC. Si, which is a nonpolar material, has a constant lattice dielectric function: $\kappa_0 = \kappa_\infty = 12.1$.

Within the RPA and including the local field correction $\bar{f}(\mathbf{q})$ proposed by Hubbard,⁸ the (test-particle–particle) dielectric function of the electron (or hole) gas $\tilde{\epsilon}(\mathbf{q}, \omega)$ is related to the polarizability $\alpha^0(\mathbf{q}, \omega)$ by^{4,5,8}

$$\tilde{\epsilon}(\mathbf{q}, \omega) = 1 + [1 - \bar{f}(\mathbf{q})] \alpha^0(\mathbf{q}, \omega) \quad (7a)$$

$$\alpha^0(\mathbf{q}, \omega) = \frac{-2}{\hbar} \frac{v(\mathbf{q})}{\kappa(\omega)} \int \frac{d\mathbf{k}}{(2\pi)^3} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi i} \times \sum_{j,j'} \Lambda_{j,j'}(\mathbf{k}, \mathbf{k}') G_j^0(\mathbf{k}, \omega') G_{j'}^0(\mathbf{k}', \omega' + \omega), \quad (7b)$$

where the summation runs over all bands which are populated by the electron gas or by the hole gas. For the present materials $j = j' = c1$ for n -type materials and $j, j' = \nu 1, \nu 2$, and $\nu 3$ for p -type materials. In order to evaluate the integral for the polarizability and get a manageable expression for $\alpha^0(\mathbf{q}, \omega)$, the energy dispersions of the Green's functions in Eq. (7b) are approximated using parabolic dispersions and setting the overlap integral to unity. Within this approximation the energy bands for calculating $\alpha^0(\mathbf{q}, \omega)$ are represented by the density-of-states effective masses m_j ($j = c1, \nu 1, \nu 2, \nu 3, \dots$).⁴ However, the uppermost valence bands in all five materials and the lowest conduction band in 6H-SiC are very nonparabolic and in order to take into account nonparabolic effects we have used concentration dependent density-of-states effective masses $m_{c1}(n)$, $m_{\nu 1}(p)$, $m_{\nu 2}(p)$, and $m_{\nu 3}(p)$, where $n(p)$ is the free electron (hole) concentration. These masses are chosen in such a way that the resulting Fermi energy of the electron gas E_F^e or of the hole gas E_F^h coincides with the Fermi energy obtained by using the numerical energy dispersion from the band structure calculation. This way of treating the nonparabolicities has in Ref. 4 been found to give accurate results. We emphasize that the parabolic approximation of the energy bands has been made only for calculating the polarizability $\alpha^0(\mathbf{q}, \omega)$; the full band structures are used otherwise in Eqs. (1)–(5).

The ν equivalent minima of the lowest conduction band in 3C-, 2H-, 4H-SiC, and Si are essentially parabolic and can accurately be described by the effective electron masses presented in Table I. The density-of-states effective mass is

$m_{c1}=[m_x m_y m_z]^{1/3}$ for these materials. In 6H-SiC the lowest conduction band is parabolic in the transverse direction, whereas in the longitudinal direction the band has a very nonparabolic double-well structure (see Sec. II C). The resulting concentration dependent density-of-states effective masses of the lowest conduction band in 6H-SiC is thus $m_{c1}(n)=[m_x m_y m_z(n)]^{1/3}$, where $m_z(n)$ is shown in Fig. 1.

For the valence bands the calculated density-of-states effective masses $m_{v1}(p)$ (solid lines), $m_{v2}(p)$ (dashed lines), and $m_{v3}(p)$ (dotted lines) are presented in Fig. 2. The hole concentration p fills the parabolic valence bands down to the Fermi energy E_F^v and the number of populated valence bands ν_p depends on the hole concentration. In 3C-SiC and Si, the energy difference between the doubly degenerate uppermost valence bands at the Γ point and the third uppermost valence band is calculated to be $E_{\nu1,\nu2}^0(\mathbf{0}) - E_{\nu3}^0(\mathbf{0}) = 14.5$ and 50.1 meV, respectively, and the onset of populating the third uppermost valence band is at about $p = 4 \times 10^{19} \text{ cm}^{-3}$ for both materials. In the hexagonal polytypes the energy differences between the uppermost and the second uppermost valence band are $E_{\nu1}^0(\mathbf{0}) - E_{\nu2}^0(\mathbf{0}) = 8.8, 8.6,$ and 8.5 meV in 2H-, 4H-, and 6H-SiC, respectively, and the second uppermost valence band starts to be populated at $p = 6 \times 10^{18} - 1 \times 10^{19} \text{ cm}^{-3}$. In 6H-SiC, the crystal-field splitting is small $[E_{\nu1}^0(\mathbf{0}) + E_{\nu2}^0(\mathbf{0})]/2 - E_{\nu3}^0(\mathbf{0}) = 53$ meV and the third uppermost valence band in 6H-SiC starts to be populated at about $2.5 \times 10^{20} \text{ cm}^{-3}$.

The local field correction $\bar{f}(\mathbf{q})$ is calculated according to Hubbard.^{4,8} For n -type materials the averaged local field correction is⁴

$$\bar{f}(\mathbf{q}) = \frac{1}{2v^2} \sum_{i=1}^v \frac{E_{c1}^0(\Delta\mathbf{k}_i + \mathbf{q})}{(E_{c1}^0(\Delta\mathbf{k}_i + \mathbf{q}) + E_F^c)}, \quad (8a)$$

$$E_F^c = \frac{\hbar^2}{2m_{c1}} (3\pi^2 n/v)^{2/3}, \quad (8b)$$

where $\Delta\mathbf{k}_i = \mathbf{k}_{m,i} - \mathbf{k}_{m,1}$ is the vector between the different equivalent minima and one specific conduction band minimum. Analogously, for p -type materials the averaged local field correction is

$$\bar{f}(\mathbf{q}) = \frac{1}{2v_p^2} \sum_{j=\nu1}^{\nu3} \frac{E_j^0(\mathbf{q}) \times \theta(E_j^0(\mathbf{0}) - E_F^v)}{(E_j^0(\mathbf{q}) + E_F^v)}, \quad (9)$$

where $\theta(x)$ is the unit step function: 1 if $x > 0$ and 0 otherwise. The Fermi energy E_F^v of the hole gas is obtained by

$$p = \frac{1}{3\pi^2} \sum_{j=\nu1}^{\nu3} \left(\frac{2m_j}{\hbar^2} [E_j^0(\mathbf{0}) - E_F^v] \times \theta(E_j^0(\mathbf{0}) - E_F^v) \right)^{3/2}. \quad (10)$$

C. Band structure and overlap integrals

The energy dispersions $E_j^0(\mathbf{k})$ of the intrinsic crystals have been calculated using a full-potential linearized augmented plane wave method, within the local density approximation to the density functional theory. A software¹⁴ program with a fully relativistic Hamiltonian and with the exchange-correlation potential of J. D. Perdew and Y. Wang¹⁵ was used. The employed band structure calculation has been found to very accurately describe the curvatures of

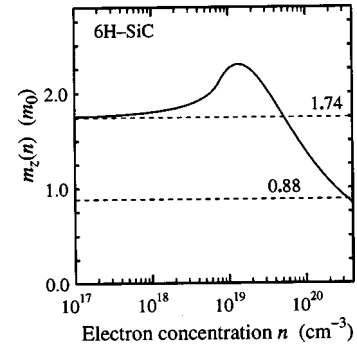


FIG. 1. Concentration dependent longitudinal effective electron mass $m_z(n)$ used in the calculation of $\alpha^0(\mathbf{q}, \omega)$ in n -type 6H-SiC. For k_z near the minimum and for large k_z the true curvature of the double-well minimum is represented by $m_z = 1.74m_0$ and $m_z = 0.88m_0$, respectively.

the lowest conduction band minima as well as the uppermost valence bands in both SiC and Si.^{6,7} Since the SiC polytypes lack inversion symmetry, the average spin independent eigenvalues were obtained by $E_j^0(\mathbf{k}) = [E_{j\uparrow}^0(\mathbf{k}) + E_{j\downarrow}^0(\mathbf{k})]/2$, where $j\uparrow$ and $j\downarrow$ are the spin-up- and spin-down-like states in the j th band. More information of the band structure calculation is to be found in Ref. 6.

For high concentrations of the electron gas, the lowest conduction band is heavily filled and the Fermi energy E_F^c of the electron gas is large (analogously for the hole gas in the valence bands). The electron and hole states near the Fermi energy are of particular importance for calculating the self-energies and, thus, the higher the concentrations the more predominant becomes the nonparabolicity of the energy bands. In particular the valence bands are very nonparabolic; in the cubic polytypes the bands are nonparabolic at the very vicinity of the Γ point and in the hexagonal polytypes the nonparabolicity is obvious for energies about 4 meV below the valence band maximum. Therefore, in the present work special attention has been paid to the effects on the self-energies due to the nonparabolicity of the energy bands.

The energy dispersion of the three uppermost valence bands have been calculated and the energy bands are represented on a mesh of about 23 000 $\mathbf{k}$ points in the first BZ, mainly concentrated about the Γ point in order to accurately describe the most important interactions between the three

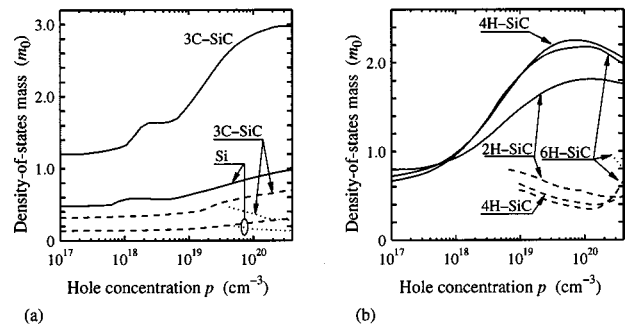


FIG. 2. The polarizability $\alpha^0(\mathbf{q}, \omega)$ in the p -type materials has been calculated using parabolic energy dispersions, based on concentration dependent density-of-states effective masses $m_{v1}(p)$ (solid lines), $m_{v2}(p)$ (dashed lines), and $m_{v3}(p)$ (dotted lines) showed here for (a) 3C-SiC and Si (b) 2H-, 4H-, and 6H-SiC.

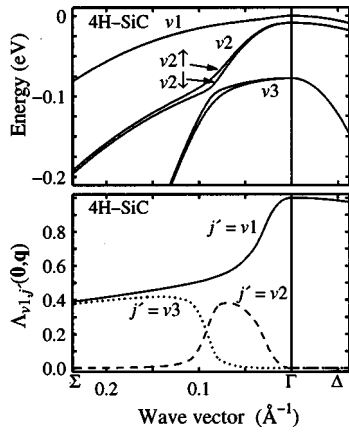


FIG. 3. The energy dispersions (upper graphs) and the overlap integrals (lower graphs) for the three uppermost valence bands in 4H-SiC in the directions $\Gamma M \equiv \Sigma$ and $\Gamma A \equiv \Delta$, starting from the Γ point and going out 20% of the total length to the BZ boundary. Calculated overlap integrals $\Delta_{v1j'}(0, \mathbf{q})$ are between initial state in the uppermost valence band and final states in the uppermost valence band ($j' = v1$, solid lines), in the second uppermost valence band ($j' = v2$, dashed lines), and in the third uppermost valence band ($j' = v3$, dotted lines).

uppermost valence bands. In Fig. 3 (upper graph) the resulting energy dispersions along two symmetry lines in 4H-SiC are shown. The nonparabolicity in the energy dispersion is evident. For calculating the energy shift of the valence band maximum the relevant overlap integrals are $\Delta_{v1j'}(0, \mathbf{q})$ [$j' = v1, v2$, and $v3$] and the overlap integrals are represented on the same mesh of $\mathbf{k}$ points as the energy bands. The resulting overlap integrals along the two symmetry lines in 4H-SiC are shown in the lower part of Fig. 3.

The necessity of using numerical overlap integrals for the valence bands is indicated in Fig. 4. In the figure the overlap integrals between states in the uppermost valence band and in the second uppermost valence band in 3C-SiC is shown, as a function of the angle θ between initial and final states $\mathbf{k}$ and $\mathbf{k}'$. The lengths of the wave vectors are $k = 0.001$ and $k' =$ (i) 0.001, (ii) 0.010, (iii) 0.050, and (iv) 0.100 in units of $(2\pi/a)$. The dotted curves represent the overlap integrals $\Delta_{v1,v1}(\theta) = 0.25 + 0.75 \times \cos^2(\theta)$ and

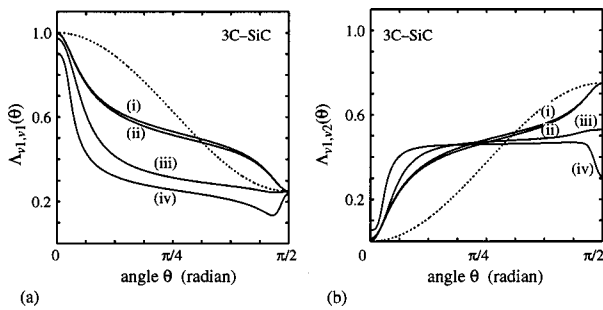


FIG. 4. Calculated overlap integral $\Delta_{v1,j'}(\theta)$ (solid lines) in 3C-SiC between initial state $\mathbf{k}$ in the uppermost valence band and final states $\mathbf{k}'$ in (a) the uppermost valence band ($j' = v1$) and in (b) the second uppermost valence band ($j' = v2$), as a function of angle θ between $\mathbf{k}$ and $\mathbf{k}'$. The lengths of the wave vectors are $k = 0.001$ and $k' =$ (i) 0.001, (ii) 0.010, (iii) 0.050, and (iv) 0.100 in units of $(2\pi/a)$. The results for Si are similar. The dotted lines are the commonly used overlap integrals $\Delta_{v1,v1}(\theta) = 0.25 + 0.75 \times \cos^2(\theta)$ and $\Delta_{v1,v2}(\theta) = 0.75 \times \sin^2(\theta)$.

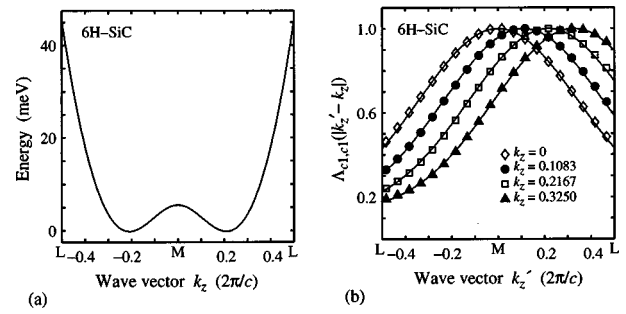


FIG. 5. (a) The energy dispersion of the lowest conduction band in 6H-SiC along the LML line and (b) the corresponding overlap integral $\Delta_{c1,c1}(k_z, k'_z)$, where the marks represent calculated values and the solid lines are the fitted values. The wave vector k_z is in units of $2\pi/c$. The energy in (a) is referenced to the conduction band minimum.

$\Delta_{v1,v2}(\theta) = 0.75 \times \sin^2(\theta)$, which are commonly used overlap integrals for cubic materials, derived by using the $\mathbf{k} \cdot \mathbf{p}$ method and excluding the nonparabolic term in the energy dispersion.¹⁶ Even for $k' = 0.001 \times (2\pi/a)$, there are large deviations between the $\mathbf{k} \cdot \mathbf{p}$ overlap integral and the overlap integral from the band structure calculations (the deviation remains when k and k' are decreased) and for large wave vectors the deviations are obvious.

The energy dispersion of the lowest conduction band $E_{c1}^0(\mathbf{k})$ in 3C-, 2H-, 4H-SiC, and Si were found to be essentially parabolic for the concentrations of interest. We therefore utilize parabolic energy dispersions for these materials, described by the three principal effective electron masses m_x , m_y , and m_z . In these materials the overlap integrals for scatterings within one and the same minimum of the lowest conduction band $\Delta_{c1,c1}(\mathbf{k}, \mathbf{k}')$ were found to be close to one for a set of $\mathbf{k}$ and $\mathbf{k}'$ important for the concentrations of interest. Furthermore, it has been shown¹¹ that intervalley scattering due to electron-impurity ion interaction between states in different equivalent minima gives only a minor contribution to the self-energies for a system of a random distribution of impurities compared to the intravalley scattering. We therefore neglect overlaps $\Delta_{c1,c1}(\mathbf{k}, \mathbf{k}')$, when $|\mathbf{k}, c1\rangle$ and $|\mathbf{k}', c1\rangle$ are states in different, but equivalent minima.

The lowest conduction band in 6H-SiC is parabolic only in the transverse direction, whereas in the longitudinal direction the band has a very nonparabolic double-well structure [Fig. 5(a)]. The band has been parametrized along k_z in Ref. 6. Due to the very flat curvature of the double-well structure and the small BZ in 6H-SiC, the Fermi wave vector reaches the BZ boundary in the k_z direction at a rather low electron concentration, namely at $n = 1.2 \times 10^{20} \text{ cm}^{-3}$ implying $E_F^c = 45 \text{ meV}$. This requires a special treatment of the overlap integral along the longitudinal direction. In Fig. 5(b), the overlap integral within the lowest conduction band minimum in 6H-SiC as a function of $k'_z = k_z + q_z$ and for different k_z is shown along the LML line. The marks are calculated values and the solid lines are from the parametrized overlap integral expression

$$\Delta_{c1,c1}(\mathbf{k}, \mathbf{k}') = \Delta_{c1,c1}(|k'_z - k_z|) = 0.1268 + 0.8732 \times e^{-4.2248 \times (q_z/(2\pi/c))^2}. \quad (11)$$

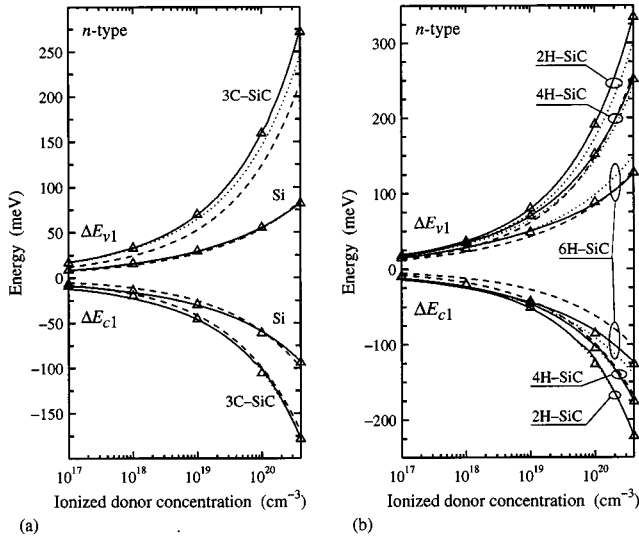


FIG. 6. Energy shift of lowest conduction band minimum $\Delta E_{c1} < 0$ and of the uppermost valence band maximum $\Delta E_{v1} > 0$ in n -type (a) 3C-SiC and Si, and (b) 2H-, 4H-, and 6H-SiC. The solid lines represent the full calculations, dotted lines are corresponding calculations with parabolic approximations of the energy dispersions and overlap integrals, and the dashed lines are the semi-empirical calculations of Ref. 18. Triangles represent Eq. (14) using the parameters in Table II.

Even when $\mathbf{k}$ and $\mathbf{k}'$ point in other directions than in the k_z direction, Eq. (11) was found to essentially represent the calculated overlap integrals and therefore this overlap integral has been used for $\Lambda_{c1,c1}(\mathbf{k}, \mathbf{k}')$ in 6H-SiC.

III. RESULTS

A. n -type SiC and Si

The doping-induced energy shifts of the lowest conduction band minimum ΔE_{c1} and of the uppermost valence band maximum ΔE_{v1} in n -type material are obtained as

$$\Delta E_{c1} = \text{Re}\{\hbar \sum_{c1}^x(\mathbf{k}_m, \zeta_{c1}^0(\mathbf{k}_m)) + \hbar \sum_{c1}^c(\mathbf{k}_m, \zeta_{c1}^0(\mathbf{k}_m)) + \hbar \sum_{c1}^{\text{ion}}(\mathbf{k}_m, \zeta_{c1}^0(\mathbf{k}_m))\}, \quad (12)$$

$$\Delta E_{v1} = \text{Re}\{\hbar \sum_{v1}^c(\mathbf{0}, \zeta_{v1}^0(\mathbf{0})) + \hbar \sum_{v1}^{\text{ion}}(\mathbf{0}, \zeta_{v1}^0(\mathbf{0}))\}, \quad (13)$$

where $\mathbf{k}_m = \mathbf{k}_X, \mathbf{k}_K, \mathbf{k}_M, \mathbf{k}_U$, and $\mathbf{k}_\Delta$ in 3C-, 2H-, 4H-, 6H-SiC, and Si, respectively. The shift of the valence band does not contain exchange contribution since the valence bands states of the undoped materials are completely occupied by electrons and the exchange interaction in the valence bands is therefore already contained in the valence band structure $E_{v1}^0(\mathbf{k})$. In Fig. 6 the energy shifts ΔE_{c1} and ΔE_{v1} as functions of ionized donor concentration N_D^+ are presented (solid lines). The dotted lines are the corresponding calculations using the parabolic approximations. Thus, in 6H-SiC the double-well structure is neglected [$m_z = 1.83m_0$, $\nu = 6$, and $\Lambda_{c1,c1}(\mathbf{k}, \mathbf{k}') = 1$] and, for all polytypes, the valence bands are represented by parabolic energy dispersions with the effective hole masses according to Table VI in Ref. 6. Moreover, in the parabolic approximation the overlap integrals for the valence bands are $\Lambda_{v1,j'}(\mathbf{0}, \mathbf{q}) = \delta_{v1,j'}$ for the hexagonal polytypes and $\Lambda_{v1,v1}(\mathbf{0}, \mathbf{q}) = [1 + 3 \cos^2(\theta)]/4$, $\Lambda_{v1,v2}(\mathbf{0}, \mathbf{q})$

$= 3 \sin^2(\theta)/4$, and $\Lambda_{v1,v3}(\mathbf{0}, \mathbf{q}) = 0$ for the cubic materials, where θ is the angle between $\mathbf{q}$ and the k_x axis.

In Fig. 6 it is obvious that inclusion of the numerical band structure will affect the self-energies. For the uppermost valence band the full calculation gives a larger energy shift than the corresponding result within the parabolic approximation, except in 6H-SiC where the double-well structure of the lowest conduction band also contributes. However, the effect is rather moderate. The reason why the parabolic approximations for Si is better than for 3C-SiC can be explained by the larger spin-orbit splitting in Si (calculated to be $\Delta_{so} = 50.1$ meV)⁷ than in 3C-SiC ($\Delta_{so} = 14.5$ meV),⁷ so that the parabolic effective hole masses in Si describe the valence band curvatures accurately in a wider energy range. Also the parabolic approximation of the overlap integrals is more accurate in Si than in 3C-SiC.

The influences of the double-well structure and of the overlap integral in 6H-SiC are obvious for ΔE_{c1} at high ionized donor concentrations. For electron concentrations larger than about $1 \times 10^{19} \text{ cm}^{-3}$ the parabolic approximation of the conduction band fails to describe the Fermi energy of the electron gas in the double-well minimum. For instance, at $n = 7.2 \times 10^{18} \text{ cm}^{-3}$ the Fermi energy of the electron gas is 5.7 meV and above this concentration the two electron gases (for $k_z > 0$ and $k_z < 0$) in the double-well minimum are not separated (see also Refs. 4 and 17).

The dashed lines in Fig. 6 for 3C-, 4H-, 6H-SiC, and Si represent the calculated self-energies of U. Lindefelt,¹⁸ in which an extended semi-empirical method originally developed by S. Jain and D. J. Roulston¹⁹ was used. Even though this model uses several approximations it rather accurately describes the energy shifts of both the lowest conduction band minimum and the uppermost valence band maximum. The calculations in Ref. 18 give smaller energy shifts than the corresponding values presented here. In particular, the energy shifts of the conduction band minimum in 6H-SiC and the valence band maximum in 3C-SiC are underestimated compared to the energy shifts calculated here.

The energy shifts of the conduction and the valence bands can efficiently be parametrized according to (see also Ref. 18)

$$\Delta E_{c1} = A_{n,c1} \left(\frac{N_D^+}{10^{18}} \right)^{1/3} + B_{n,c1} \left(\frac{N_D^+}{10^{18}} \right)^{1/2}, \quad (14a)$$

$$\Delta E_{v1} = A_{n,v1} \left(\frac{N_D^+}{10^{18}} \right)^{1/4} + B_{n,v1} \left(\frac{N_D^+}{10^{18}} \right)^{1/2}. \quad (14b)$$

The fitted parameters $A_{n,c1}$, $B_{n,c1}$, $A_{n,v1}$, and $B_{n,v1}$ are presented in Table II. To indicate the validity of this parametrization, we show in Fig. 6 (triangles) the energy shifts using Eq. (14).

Since ΔE_{c1} is a negative and ΔE_{v1} is a positive quantity the total energy shift of the fundamental band gap ΔE_g is

$$\Delta E_g = \Delta E_{c1} - \Delta E_{v1} \quad (15)$$

and the fundamental band gap is given by $E_g = E_g^0 + \Delta E_g$.

The corresponding optical band gap in the present n -type materials is defined as the energy needed to promote one

TABLE II. Parameters describing ΔE_{c1} and ΔE_{v1} [Eq. (14)] in *n*-type materials.

<i>n</i> -type (meV)	Si	3C-SiC	2H-SiC	4H-SiC	6H-SiC
$A_{n,c1}$	-14.84	-17.22	-15.86	-17.91	-23.31
$B_{n,c1}$	0.78	-2.57	-5.25	-2.20	2.26
$A_{n,v1}$	15.08	25.05	25.58	28.23	26.57
$B_{n,v1}$	0.74	8.01	11.01	6.24	0.43

electron from the top of the valence band to the Fermi energy of the electron gas. The doping-induced energy shift of the optical band gap can be estimated by

$$\Delta E_g^{\text{opt}} = \Delta E_g + E_F^c. \quad (16)$$

Here, it is presumed that the curvature of the conduction band is not affected by the doping. This approximation has been found to be reasonable.⁴ In Fig. 7 the resulting energy shifts of the fundamental band gap (solid lines) and of the optical band gap (dashed lines) are shown. The marks in Fig. 7(a) indicate measured energy shifts of the fundamental band gap^{20–22} (open circles, triangles, squares, and plus signs) and of the optical band gap²⁰ (filled circles) in Si. There are fairly good agreements between calculated and measured values of both ΔE_g and ΔE_g^{opt} .

B. *p*-type SiC and Si

The doping-induced energy shifts of the lowest conduction band minimum ΔE_{c1} and of the uppermost valence band maximum ΔE_{v1} in *p*-type materials are, according to the arguments in the previous sections, obtained as

$$\Delta E_{c1} = \text{Re}\{-\hbar \sum_{c1}^c(\mathbf{k}_m, -\zeta_{c1}^0(\mathbf{k}_m)) - \hbar \sum_{c1}^{\text{ion}}(\mathbf{k}_m, -\zeta_{c1}^0(\mathbf{k}_m))\}, \quad (17)$$

$$\Delta E_{v1} = \text{Re}\{-\hbar \sum_{v1}^x(\mathbf{0}, -\zeta_{v1}^0(\mathbf{0})) - \hbar \sum_{v1}^c(\mathbf{0}, -\zeta_{v1}^0(\mathbf{0})) - \hbar \sum_{v1}^{\text{ion}}(\mathbf{0}, -\zeta_{v1}^0(\mathbf{0}))\}. \quad (18)$$

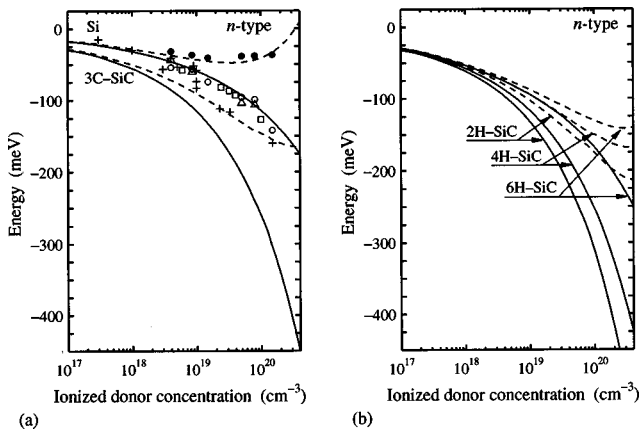


FIG. 7. The energy shifts of the fundamental band gap (solid lines) and of the optical band gap (dashed lines) in *n*-type (a) 3C-SiC and Si, and (b) 2H-, 4H-, and 6H-SiC. Open circles [Ref. 20], triangles [Ref. 20], squares [Ref. 21], and plus signs [Ref. 22] are measured values of ΔE_g for Si and filled circles [Ref. 20] are corresponding values for ΔE_g^{opt} .

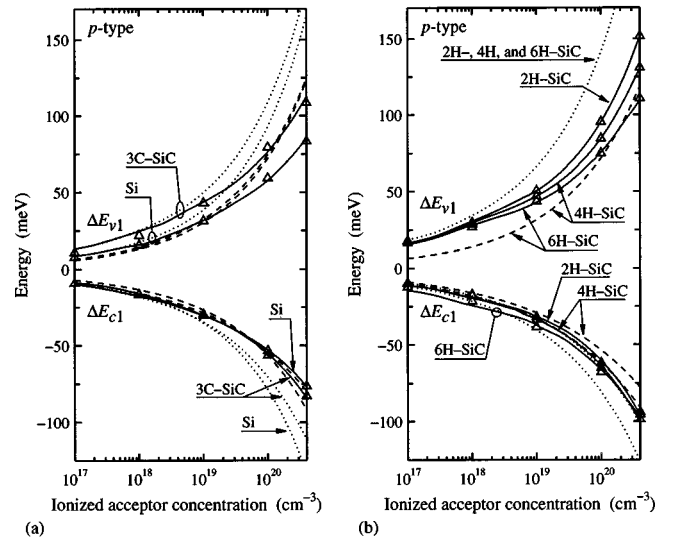


FIG. 8. Energy shift of lowest conduction band minimum $\Delta E_{c1} < 0$ and of the uppermost valence band maximum $\Delta E_{v1} > 0$ in *p*-type (a) 3C-SiC and Si, and (b) 2H-, 4H-, and 6H-SiC. The solid lines represent the full calculations, dotted lines are corresponding calculations with parabolic approximations of the energy dispersions and overlap integrals, and the dashed lines are semi-empirical calculations of Ref. 18. Triangles represent Eq. (19) using the parameters in Table III.

In Fig. 8, the energy shifts ΔE_{c1} and ΔE_{v1} as a function of ionized acceptor concentration N_A^- are presented (solid lines). The dotted lines represent the calculation within the parabolic approximation. For low ionized acceptor concentrations the parabolic approximation gives a good description of the energy shifts. This is what could be expected, since the employed crystal parameters for the parabolic approximation were chosen to describe the energy dispersion at the very top of the valence bands. For high hole concentrations, however, the parabolic approximation of the valence bands fails to represent the true Fermi energy and the resulting energy shifts ΔE_{c1} and ΔE_{v1} differ from the results obtained by using the numerical energy dispersions. Also the double-well structure of the lowest conduction band in 6H-SiC has an important effect on ΔE_{c1} for high ionized acceptor concentrations.

The details in the curvature of ΔE_{v1} for the hexagonal polytypes can be explained directly from the energy dispersion of the three valence bands (see Fig. 3). For hole concentrations $p = 1 \times 10^{17} - 1 \times 10^{18} \text{ cm}^{-3}$ only the very top of the uppermost valence band is populated by holes and in this energy region ($E_F^v > -4 \text{ meV}$) the band is essentially parabolic and, therefore, the parabolic approximation is good for these concentrations. At a hole concentration of $1 \times 10^{18} \text{ cm}^{-3}$ the Fermi energy is about -4 meV in all three hexagonal polytypes and at this energy the uppermost valence band becomes nonparabolic and thus the parabolic approximation becomes inaccurate. At the hole concentrations $p = 6, 9, \text{ and } 9 \times 10^{18} \text{ cm}^{-3}$ in 2H-, 4H-, and 6H-SiC, respectively, the second uppermost valence bands start to be populated. Here, the effect on ΔE_{v1} is most prominent in 2H-SiC, because 2H-SiC has the lowest concentration for the onset of populating the second uppermost valence band and,

TABLE III. Parameters describing ΔE_{c1} and ΔE_{v1} [Eq. (19)] in p -type materials.

p -type (meV)	Si	3C-SiC	2H-SiC	4H-SiC	6H-SiC
$A_{p,c1}$	-16.27	-16.18	-16.93	-16.15	-21.77
$B_{p,c1}$	-0.18	-0.52	-1.14	-1.07	0.10
$A_{p,v1}$	18.46	26.36	-33.22	-35.07	-27.88
$B_{p,v1}$	-2.63	-4.28	7.41	6.74	4.60
$C_{p,v1}$			55.53	56.96	50.13

moreover, the density-of-states effective masses $m_{v1}(p)$ and $m_{v2}(p)$ are smaller and larger, respectively, in 2H-SiC (see Fig. 2) than the corresponding masses in the other two hexagonal polytypes. This results in a larger energy shift ΔE_{v1} for 2H-SiC than for 4H- and 6H-SiC.

The dashed lines in Fig. 8 are from the semi-empirical model of Ref. 18. These calculations give roughly the same energy shifts of the conduction bands as the present calculations, but the disagreements in the calculated energy shifts of the valence bands are more pronounced, especially in the hexagonal polytypes at low ionized acceptor concentrations. The main reason for this is probably that the employed effective hole mass ($m_{v1}=1.70m_0$) for the hexagonal polytypes in Ref. 18 represents the true energy dispersion for large wave vectors, but inaccurately describes the energy dispersion at the valence band maximum which is the most important region for low concentrations.

The triangles in Fig. 8 represent the fitted energy shifts according to

$$\Delta E_{c1} = A_{p,c1} \left(\frac{N_A^-}{10^{18}} \right)^{1/4} + B_{p,c1} \left(\frac{N_A^-}{10^{18}} \right)^{1/2}, \quad (19a)$$

$$\Delta E_{v1} = A_{p,v1} \left(\frac{N_A^-}{10^{18}} \right)^{1/3} + B_{p,v1} \left(\frac{N_A^-}{10^{18}} \right)^{1/2} + C_{p,v1} \left(\frac{N_A^-}{10^{18}} \right)^{1/4}, \quad (19b)$$

where the resulting parameters are presented in Table III. In Eq. (19b) the third term has been added in order to improve the fit.

The narrowing of the fundamental band gap $\Delta E_g = \Delta E_{c1} - \Delta E_{v1}$ and the shift of the optical band gap $\Delta E_g^{\text{opt}} = \Delta E_g - E_F^v$ are shown in Fig. 9. The rather small differences in the energy shifts of both the fundamental band gap and the optical band gap between the polytypes 2H-, 4H-, and 6H-SiC are a consequence of the similar valence band structures for the hexagonal polytypes. The marks indicate measured energy shifts²⁰ of the fundamental band gap (open circles and triangles) and of the optical band gap (filled circles) in Si; the calculated and measured values of ΔE_g agree well, whereas calculated shift ΔE_g^{opt} is somewhat overestimated. The dotted line in Fig. 9(a) represents the calculated shift ΔE_g in Si using the parabolic approximation. It is obvious that the nonparabolic treatment of the valence bands results in much better agreement between calculated and measured values at high acceptor concentrations. The parabolic approximation strongly overestimates the narrowing of the fundamental band gap.

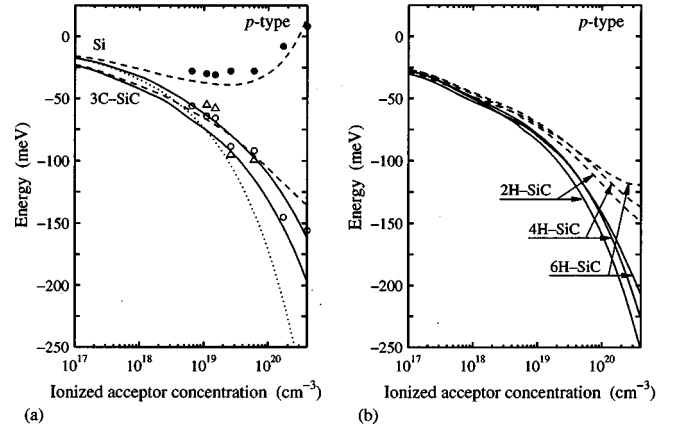


FIG. 9. The energy shifts of the fundamental band gap (solid lines) and of the optical band gap (dashed lines) in p -type (a) 3C-SiC and Si, and (b) 2H-, 4H-, and 6H-SiC. The dotted line in (a) represents calculated shift ΔE_g for Si using the parabolic approximation. Open circles and triangles are measured values of ΔE_g for Si and filled circles are corresponding values for ΔE_g^{opt} [Ref. 20].

IV. SUMMARY

The doping induced energy shifts of the lowest conduction band minimum and the uppermost valence band maximum have been calculated for both n -type and p -type 3C-, 2H-, 4H-, 6H-SiC, and Si, using the RPA with the local field correction of Hubbard. The results show that including the nonparabolicity of the energy dispersion of the valence bands in all five materials and the double-well structure of the conduction band in 6H-SiC has a strong impact on the resulting self-energies both for the valence and for the conduction band states. The reduction of the band gap narrowing due to the nonparabolicity is large in p -type SiC and Si. Similar results have been reported for p -type GaAs,²³ and better agreement between theory and experiments was obtained when the nonparabolicity of the valence bands was included in the calculations.

The method of calculation was based on a zero-temperature formalism and the results are in a strict sense valid only for impurity concentration above the concentration for the metal nonmetal transition, which is about $4 \times 10^{18} \text{ cm}^{-3}$ in n -type Si and 3C-SiC and about $0.6\text{--}3 \times 10^{19} \text{ cm}^{-3}$ in n -type 4H- and 6H-SiC.⁴ For n -type Si a finite-temperature RPA calculation has been performed by B. E. Sernelius²⁴ and by F. Thuselt and M. Rösler,²⁵ showing that the zero-temperature calculation is a good approximation for calculations of ΔE_{c1} even at finite temperatures, mainly due to compensating effects between the correlation and the exchange interactions. However, in n -type (p -type) materials ΔE_{v1} (ΔE_{c1}) does not depend on the exchange energy and this shift might therefore be more affected by the temperature. Nevertheless, it is believed that the zero-temperature calculations presented here can, with reasonable accuracy, be applied for finite temperatures for concentrations below the Mott transition where the impurities are partly or fully thermally ionized. The temperature dependent carrier concentration has been worked out for Al-doped 4H-SiC and N-doped 6H-SiC by Persson and Lindefelt.²⁶

The calculations took into account the nonparabolicities of the energy bands as well as the temperature dependence of the intrinsic band gap E_g^0 , but neglected the possibility for the metal nonmetal transition.

In Ref. 4, the $\mathbf{k}$ dependence of the self-energy has been investigated for n -type SiC and Si. The calculations show that the energy bands are somewhat distorted due to the doping-induced self-energy. For example, the double-well structure in 6H-SiC almost disappears for a donor concentration $5 \times 10^{19} \text{ cm}^{-3}$.

In Eq. (2), Sec. II A, the umklapp processes are neglected because it is here assumed that the Fermi wave vector is small compared to the length of the BZ, and only the wave vectors near the Fermi vector predominate in the contribution to the self-energy. However, since the BZ along the c axis in 6H-SiC is small, this approximation might be inaccurate for very high impurity concentrations in 6H-SiC. For instance, the conduction band in 6H-SiC is populated up to the BZ boundary in the k_z direction at the electron concentration $n = 1.2 \times 10^{20} \text{ cm}^{-3}$.

ACKNOWLEDGMENTS

This work was financially supported by the Swedish Research Council for Technical Sciences and by the Swedish Natural Science Research Council.

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